课程笔记 SNN训练算法汇总综述

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摘要

课程笔记 SNN训练算法汇总综述，涵盖了从传统的 BPTT/STBP 到最新的在线学习方法（OTTT、NDOT）以及局部学习规则（S-TLLR、TESS）。重点分析了每种方法的核心思想、数学推导、优缺点以及适用场景。通过对比不同算法在时间依赖性处理、梯度计算复杂度和生物可解释性方面的差异，全面总结了 SNN 训练领域的研究进展和未来方向。

PPT `Week4_SNN Training.pptx`

文章汇总

RTRL: **A learning algorithm for continually running fully recurrent neural networks** [1];

E-prop: **A solution to the learning dilemma for recurrent networks of spiking neurons** [2];

OTTT: **Online Training Through Time for Spiking Neural Networks** [3];

NDOT: **NDOT: Neuronal Dynamics-based Online Training for Spiking Neural Networks** [4];

S-TLLR: **S-TLLR: STDP-inspired Temporal Local Learning Rule for Spiking Neural Networks** [5];

TESS: **TESS: A Scalable Temporally and Spatially Local Learning Rule for Spiking Neural Networks** [6]

I. Eligibility Trace: Replacing BPTT Memory

Eligibility Trace: Replacing BPTT Memory

The Bottleneck of BPTT:

- Unrolls the computational graph over T time steps.
- Memory complexity: $\mathcal{O}(T \times n)$ (stores all states).
- Prevents online / on-device learning.

Core Idea of Eligibility Trace:

- Replace historical memory with a *recursive state*.
- Accumulates past effects into a local "trace" e_{ji}^t.
- Enables *forward-in-time* gradient computation.

Unified Mathematical Form:

$$\Delta W_{ji} = \sum_t L_j^t \cdot e_{ji}^t \tag{1}$$

- L_j^t: Top-down learning signal (error/reward).

- e_{ji}^t: Eligibility trace (local synaptic history).

Why BPTT is Memory-Intensive:

- Forward pass: $u^{(l)}[t] = f(u^{(l)}[t-1], \dots)$ must store all $u^{(l)}[t]$ for all l, t.
- Backward pass: $\frac{\partial L}{\partial u^{(l)}[t]}$ depends on future time steps, requiring reverse traversal.
- Total stored states: $\sum_{l=1}^{L} \sum_{t=1}^{T} \dim(u^{(l)}[t]) = O(TLn)$.

Derivation of the Unified Form: From the chain rule over time and space:

$$\frac{\partial L}{\partial W_{ji}} = \sum_t \frac{\partial L}{\partial y_j[t]} \frac{\partial y_j[t]}{\partial u_j[t]} \frac{\partial u_j[t]}{\partial W_{ji}} \tag{2}$$

$$= \sum_t \left(\frac{\partial L}{\partial y_j[t]} \frac{\partial y_j[t]}{\partial u_j[t]} \right) \cdot \left(\sum_{\tau \le t} \frac{\partial u_j[t]}{\partial u_j[\tau]} \frac{\partial u_j[\tau]}{\partial W_{ji}} \right) \tag{3}$$

$$= \sum_t L_j^t \cdot e_{ji}^t, \tag{4}$$

where e_{ji}^t absorbs all temporal dependencies up to time t.

讲稿对应本页PPT：

”本页介绍在线学习算法的核心动机——替代 BPTT 的内存瓶颈。BPTT 需要展开所有时间步并存储中间状态，内存为 $O(Tn)$。Eligibility Trace 的核心思想是用一个递归状态替代完整历史记忆。统一数学形式为 $\Delta W = \sum L \cdot e$，其中 L 是顶向下学习信号，e 是局部资格迹。

新增内容：我们展示了 BPTT 为何内存密集——前向需要存储所有时间步的膜电位，反向需要从未来向过去传播。通过链式法则，将梯度分解为瞬时误差项 L_j^t 和累计历史项 e_{ji}^t，后者正是 eligibility trace 的由来。”

II. Represent Work—RTRL: Real-Time Recurrent Learning

RTRL: Real-Time Recurrent Learning (1989)

Key Innovation: Computes *exact* gradients online (no unrolling).

Gradient Sensitivity Dynamics:

$$p_{ij}^k(t+1) = f_k'(s_k(t)) \left[\sum_{l \in U} w_{kl} p_{ij}^l(t) + \delta_{ik} z_j(t) \right] \tag{5}$$

- $p_{ij}^k(t) \triangleq \frac{\partial y_k(t)}{\partial w_{ij}}$: Sensitivity.
- $\sum_l w_{kl} p_{ij}^l(t)$: **Recursive eligibility** (past influences).
- $\delta_{ik} z_j(t)$: Immediate presynaptic input.

Weight Update:

$$\Delta w_{ij}(t) = \alpha \sum_{k \in U} e_k(t) p_{ij}^k(t) \tag{6}$$

Limitation: Memory Complexity: $\mathcal{O}(n^3)$ (impractical for large nets).

Derivation from the Chain Rule:

- Starting from $\frac{\partial y_k(t+1)}{\partial w_{ij}} = f'_k(s_k(t)) \left(\sum_l w_{kl} \frac{\partial y_l(t)}{\partial w_{ij}} + \delta_{ik} z_j(t) \right)$.
- The term $\sum_l w_{kl} \frac{\partial y_l(t)}{\partial w_{ij}}$ propagates the sensitivity from all neurons l at time t to neuron k at $t+1$.
- This is exactly a forward-in-time recursion: no need to store past gradients, only the current $p_{ij}^k(t)$.

Why $\mathcal{O}(n^3)$?

- For each weight w_{ij} (there are $\mathcal{O}(n^2)$), we need to store and update n sensitivities p_{ij}^k for all k.
- Total: $\mathcal{O}(n) \times \mathcal{O}(n^2) = \mathcal{O}(n^3)$.
- Also time complexity per step: $\mathcal{O}(n^4)$ if naively updating all p's.

讲稿对应本页PPT：

”RTRL 是 Williams 和 Zipser 于 1989 年提出的开山之作，核心创新是首次实现了精确梯度的在线计算。公式中 p_{ij}^k 表示输出对权重的敏感度。其递归项 $\sum w \cdot p$ 将过去影响向前传递，构成资格迹的雏形。权重的更新为误差 e_k 乘以敏感度 p。但其致命局限是内存复杂度高达 $O(n^3)$，无法用于大规模网络。

新增内容：我们从链式法则推导了该递归方程，解释了为何这是前向计算——只需当前敏感度，无需存储历史。复杂度分析表明，对于每个权重需要存储 n 个敏感度，因此总体为 $O(n^3)$，时间上为 $O(n^4)$，这解释了为何 RTRL 难以实用。”

III. Represent Work—e-Prop

e-Prop: Eligibility Propagation (2020)

Key Innovation: Biologically plausible approximation using neuron-specific feedback.

Fundamental Factorization:

$$\frac{dE}{dW_{ji}} = \sum_t L_j^t \cdot e_{ji}^t \tag{7}$$

- $L_j^t \triangleq \frac{\partial E}{\partial z_j^t}$: Learning signal.
- $e_{ji}^t \triangleq \left[\frac{dz_j^t}{dW_{ji}} \right]_{\text{local}}$: Eligibility trace.

Eligibility Trace for ALIF (Adaptive) Neurons:

$$e_{ji}^t = \psi_j^t \left(\bar{z}_i^{t-1} - \beta \epsilon_{ji,a}^t \right) \tag{8}$$

- ψ_j^t: Surrogate gradient (pseudo-derivative).
- $\epsilon_{ji,a}^t$: Slow component from threshold adaptation.

Key Insight: Slow hidden variables provide “highways into the future” for gradients.

How e-prop factorizes BPTT:

- BPTT: $\frac{dE}{dW_{ji}} = \sum_{t'} \frac{dE}{d\mathbf{h}_j^{t'}} \cdot \frac{\partial \mathbf{h}_j^{t'}}{\partial W_{ji}}$.

- Expand $\frac{dE}{d\mathbf{h}_j^{t'}}$ recursively:

$$\frac{dE}{d\mathbf{h}_j^{t'}} = \frac{dE}{dz_j^{t'}}\frac{\partial z_j^{t'}}{\partial \mathbf{h}_j^{t'}} + \frac{dE}{d\mathbf{h}_j^{t'+1}}\frac{\partial \mathbf{h}_j^{t'+1}}{\partial \mathbf{h}_j^{t'}}.$$

- By repeated substitution and swapping sums, we get:

$$\frac{dE}{dW_{ji}} = \sum_t \frac{dE}{dz_j^t} \cdot \underbrace{\left(\frac{\partial z_j^t}{\partial \mathbf{h}_j^t}\sum_{\tau \leq t}\frac{\partial \mathbf{h}_j^t}{\partial \mathbf{h}_j^\tau}\frac{\partial \mathbf{h}_j^\tau}{\partial W_{ji}}\right)}_{e_{ji}^t}. \tag{9}$$

- This re-factoring pushes all temporal dependencies into the eligibility trace e_{ji}^t, which can be computed forward in time.

Three Variants:

- *Symmetric e-prop*: $B_{jk} = W_{kj}^{\text{out}}$ (weight sharing).
- *Random e-prop*: B_{jk} fixed random (Broadcast Alignment).
- *Adaptive e-prop*: B_{jk} learned via local plasticity.

讲稿对应本页PPT：

”e-Prop 由 Bellec 等人于 2020 年提出，核心是提供了生物合理的 BPTT 近似。其基础分解式为 $dE/dW = \sum L \cdot e$。针对具有自适应阈值的 ALIF 神经元，其资格迹 e 包含代理梯度 ψ 和慢变量 ϵ。这个慢变量来自阈值适应，能将梯度信息在长时间尺度上向前传播，充当了通往未来的’高速公路’，这是 e-Prop 在生物合理性上的关键洞察。

新增内容：我们展示了 e-prop 如何通过重新因子化 BPTT 的梯度表达式，将所有时间依赖都吸收进资格迹 e_{ji}^t，使其能在线计算。此外，e-prop 有三种变体（对称、随机、自适应），随机变体甚至不需要精确的误差反向传播，进一步增强了生物可解释性。”

IV. Represent Work—OTTT (Online Training Through Time)

OTTT: Online Training Through Time (2022)

Key Innovation: Decouples temporal dependency to achieve $\mathcal{O}(n)$ memory.

Decoupled Gradient Formulation:

$$\nabla_{\mathbf{W}^l} L[t] = \mathbf{g}_{\mathbf{u}^{l+1}}[t] \cdot \left(\hat{\mathbf{a}}^l[t]\right)^\top \tag{10}$$

- $\mathbf{g}_{\mathbf{u}^{l+1}}[t]$: Instantaneous spatial gradient.
- $\hat{\mathbf{a}}^l[t]$: Presynaptic activity trace (temporal gradient).

Recursive Trace Update:

$$\hat{a}^l[t+1] = \lambda \hat{a}^l[t] + s^l[t+1] \tag{11}$$

- Tracks exponentially decaying sum of presynaptic spikes.
- Replaces expensive $\prod \frac{\partial u}{\partial u}$ terms.

Theoretical Guarantee:

- OTTT gradients provide a valid *descent direction* for optimization.

Why OTTT Works (Sketch of Theorem 1):

- Let $\mathbf{a}^l[t]$ be the weighted firing rate of layer l up to time t.
- Under mild assumptions (convergent inputs, small errors), OTTT gradient $\nabla_{\mathbf{W}^l} L$ satisfies:

$$\langle \nabla_{\mathbf{W}^l} L, (\nabla_{\mathbf{W}^l} L_{sr})_{sr} \rangle > 0, \tag{12}$$

where $(\nabla_{\mathbf{W}^l} L_{sr})_{sr}$ is the gradient computed by the spike-representation method (which is known to be a descent direction).

- This proves that OTTT, despite being an approximation, does not hurt convergence.

Complexity Comparison:

Method	Memory	Time per step
BPTT	$O(Tn)$	$O(Tn^2)$
OTTT	$O(n)$	$O(n^2)$

讲稿对应本页PPT：

”OTTT 由 Xiao 等人于 2022 年提出，核心创新是将时间依赖解耦，实现 $O(n)$ 常数内存。其解耦梯度为空间梯度 g 乘以时间梯度 $\hat{a}$。$\hat{a}$ 的递归更新为 $\hat{a}[t+1] = \lambda \hat{a}[t] + s[t+1]$，即指数衰减的突触前活动之和，替代了 BPTT 中复杂的连乘项。此外，OTTT 从理论上证明了其梯度能提供有效的下降方向。

新增内容：我们简述了定理的证明思路——通过将 OTTT 梯度与已知的下降方向（基于脉冲表示）比较，证明了其内积为正，从而保证了收敛性。复杂度对比表清晰展示了 OTTT 在内存上的巨大优势。”

V. Represent Work—NDOT (Neuronal Dynamics-based Online Training)

NDOT: Neuronal Dynamics-based Online Training (2024)

Key Innovation: Uses *continuous* neuronal dynamics for precise temporal dependency.

Continuous Temporal Dependency (Derivative Ratio):

$$e^l[t] \triangleq \frac{\partial \mathbf{u}^l[t+1]}{\partial \mathbf{u}^l[t]} = \frac{\mathbf{u}^l[t] - V_{th}\mathbf{s}^l[t]}{\mathbf{u}^l[t-1] - V_{th}\mathbf{s}^l[t-1]} \tag{13}$$

- Derived from LIF dynamics: $e(t) = u'(t+1) \oslash u'(t)$.
- $\oslash$ denotes element-wise division.

Precise Temporal Gradient Tracking:

$$\hat{\mathbf{a}}^{l-1}[t] = \mathbf{e}^{l-1}[t-1] \odot \hat{\mathbf{a}}^{l-1}[t-1] + \mathbf{s}^{l-1}[t] \tag{14}$$

- Uses dynamic $\mathbf{e}^{l-1}[t-1]$ (vs fixed λ in OTTT).

- Achieves higher accuracy via exact history modeling.

Derivation of the Continuous Ratio:

- Continuous LIF: $\tau \frac{du}{dt} = -(u - u_{\text{rest}}) + I(t)$.
- For two consecutive times t and $t + \Delta t$, we have:

$$\frac{du(t+\Delta t)}{dt} \Big/ \frac{du(t)}{dt} = \frac{u(t+\Delta t) - I(t+\Delta t)}{u(t) - I(t)} + O(\Delta t). \tag{15}$$

- In the limit $\Delta t \to 0$, this gives $e(t) = u'(t+1)/u'(t)$.
- Discretizing (with reset $u[t+1] = \lambda(u[t] - V_{th}s[t]) + I[t+1]$), we obtain the exact ratio:

$$e^l[t] = \frac{u^l[t] - V_{th}s^l[t]}{u^l[t-1] - V_{th}s^l[t-1]}. \tag{16}$$

Why NDOT outperforms OTTT:

- OTTT uses a fixed λ (e.g., 0.5) ignoring neuron state variations.
- NDOT uses the actual ratio of membrane potentials, adapting to neuronal dynamics.
- This leads to more accurate temporal credit assignment, especially for long sequences.

讲稿对应本页PPT：

”NDOT 由 Jiang 等人于 2024 年提出，核心是使用神经元动力学的连续形式计算精确的时间依赖。公式中 $e^l[t]$ 定义为膜电位对时间的导数之比，源自 LIF 连续动力学。与传统 OTTT 使用固定衰减 λ 不同，NDOT 使用动态的 e 来更新时间梯度 $\hat{a}$，这使得历史建模更精确，在复杂数据集上取得了更高的准确率。

新增内容：我们详细推导了连续比例式，展示了如何从连续 LIF 方程得到离散的 $e^l[t]$。这一动态衰减因子反映了神经元实际状态，而非固定常数，因此 NDOT 能更好地处理长时依赖。”

VI. Represent Work—S-TLLR: STDP-inspired Temporal Local Learning Rule

S-TLLR: STDP-inspired Temporal Local Learning (2025)

Key Innovation: Incorporates both *causal* and *non-causal* (STDP) spike-timing relations.

Generalized STDP Eligibility Trace:

$$e_{ij}[t] = \alpha_{pre}\Psi(u_i[t]) \cdot \text{tr}(x_j)[t] + \alpha_{post}x_j[t] \cdot (\text{tr}(\Psi(u_i[t])) - \Psi(u_i[t])) \tag{17}$$

- **Causal term** (α_{pre}): Post fires *after* pre.
- **Non-causal term** (α_{post}): Post fires *before* pre.

Task-dependent Preference: **Advantage:** Memory $\mathcal{O}(n)$; leverages relations overlooked by BPTT.

Dataset	Optimal α_{post}	Temporal Nature
DVS Gesture	-1 (Causality)	Spatial-dominant
SHD	$+1$ (Synchrony)	Temporal-dominant

From STDP to Eligibility Trace:

- Original STDP: $\Delta w_{ij} = \begin{cases} A_+ e^{-(t_i - t_j)/\tau_+}, & t_i \geq t_j \quad \text{(LTP)} \\ -A_- e^{-(t_j - t_i)/\tau_-}, & t_i < t_j \quad \text{(LTD)} \end{cases}$
- In S-TLLR, we replace exact spike times with traces (low-pass filters of spikes).
- The causal term uses presynaptic trace $\mathrm{tr}(x_j)[t]$, the non-causal term uses postsynaptic trace $\mathrm{tr}(\Psi(u_i))$.
- This yields the eligibility trace formula above, which is fully local in time.

Why Non-Causal Matters:

- Traditional BPTT only propagates gradients forward in time, thus only causal relations are considered.
- S-TLLR allows weights to be strengthened even if post fires before pre (synchrony), which can be beneficial for tasks where temporal order is less critical.
- The experiments show that the optimal α_{post} sign depends on the dataset's temporal characteristics.

讲稿对应本页PPT：

”S-TLLR 由 Apolinario 和 Roy 于 2025 年提出，核心创新是引入了 STDP 中的非因果项。其资格迹包含两项：因果项（前发放后，α_{pre}）和非因果项（后发放前，α_{post}）。实验表明，在空间主导的视觉任务中，负的 α_{post}（强调因果）效果更好；而在时间主导的音频任务中，正的 α_{post}（强调同步）效果更好。这揭示了 BPTT 忽略的非因果关系在特定任务中的重要性。

新增内容：我们回顾了原始 STDP 方程，并解释了如何将其转化为基于迹的资格迹形式。非因果项的引入让 S-TLLR 能够利用传统 BPTT 无法捕捉的时序关系，这对某些任务（如需要同步检测的音频分类）可能更有优势。”

VII. Represent Work—TESS: Temporally and Spatially Local Learning Rule

TESS: Temporally & Spatially Local Learning (2025)

Key Innovation: Achieves *full locality* (time & space) via local learning signals.

Time Locality (Eligibility Traces):

$$e^l_{pre}[t] = \alpha_{pre}\Psi(u^l[t]) \otimes q^l[t], \quad e^l_{post}[t] = \alpha_{post} h^l[t] \otimes o^{l-1}[t] \tag{18}$$

Space Locality (Local Signal Generator):

$$m^l[t] = B^{l^\top}\left(f(B^l o^l[t]) - y^*\right) \tag{19}$$

- B^l: Fixed quasi-orthogonal binary matrix (square waves).
- $f(\cdot)$: Softmax (classification) or Identity.
- No backpropagation across layers: $\mathcal{O}(Cn)$ vs $\mathcal{O}(n^2)$.

Performance: Matches BPTT on CIFAR10/DVS Gesture with $205\times$ fewer MACs.

Complexity Comparison: Why TESS works:

- The fixed binary matrix B^l acts as a set of random projections that preserve task-relevant information.

Method	Memory	Time (per step)	Spatial Locality?
BPTT	$O(TLn)$	$O(TLn^2)$	No
S-TLLR	$O(Ln)$	$O(Ln^2)$	No
TESS	$O(Ln)$	$O(LCn)$	Yes

- The square-wave basis ensures quasi-orthogonality and hardware efficiency.
- Each layer independently computes its own error signal, eliminating the need for error backpropagation.
- This design is inspired by neural activity synchronization in biological systems.

讲稿对应本页PPT：

”TESS 由 Apolinario 等人于 2025 年提出，实现了时间和空间的完全局部学习。时间局部性继承自 S-TLLR 的因果和非因果迹。空间局部性是其最大突破：使用固定的准正交二进制矩阵 B 将本层输出投影到任务子空间，与标签比较后投影回本地，生成学习信号 m，完全不需要跨层反向传播。计算复杂度从 $O(n^2)$ 降至 $O(Cn)$，在保持 BPTT 精度的同时，MAC 操作减少了 200 倍以上。

新增内容：我们给出了更全面的复杂度对比表，显示 TESS 在时间和空间上都是局部的，且时间复杂度仅依赖于类别数 C（远小于神经元数 n）。其设计动机源于生物神经元的同步活动，这为完全局部学习提供了理论基础。”

VIII. Discussion Questions

Discussion Questions

1) **Exact vs. Approximate:** RTRL gives exact gradients ($\mathcal{O}(n^3)$). OTTT/TESS approximate for efficiency. When would you sacrifice memory for exactness, and vice versa?
2) **Biological Plausibility:** E-prop uses “slow variables” (ALIF). S-TLLR uses “non-causal STDP”. How do these bio-mechanisms offer computational advantages over standard BPTT?
3) **Future of Local Learning:** TESS eliminates global backpropagation via random projections. Can fully local rules eventually surpass BPTT in generalization, or are they inherently limited by lack of global coordination?

Additional Points for Discussion:

- *Trade-offs in practice:* While TESS reduces MACs drastically, it may require more epochs to converge. How to balance convergence speed and resource usage?
- *Hybrid approaches:* Could we combine the temporal accuracy of NDOT with the spatial locality of TESS to get the best of both worlds?
- *Hardware implications:* Which algorithm is most suitable for on-chip learning (e.g., Loihi, SpiNNaker)? Consider memory, computation, and communication overhead.

讲稿对应本页PPT：

”讨论环节包含三个开放性问题。第一，精确与近似的权衡：何时该选择 RTRL 的精确梯度，何时该选择 OTTT 的效率？第二，生物合理性：E-prop 的慢变量和 S-TLLR 的非因果 STDP 如何

提供超越标准 BPTT 的计算优势？第三，局部学习的未来：TESS 这类完全局部算法能否在泛化性上最终超越 BPTT，还是受限于缺乏全局协调？

新增讨论点：我们还提出了一些实践中的权衡，如收敛速度与资源消耗的平衡，是否可能混合不同算法的优点，以及各种算法在现有神经形态硬件上的适用性。”

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